Akshay Arjun

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Education

BITS Pilani, Hyderabad *2022 – 2026*
B.E. (Hons.) Electronics and Instrumentation Engineering *CGPA: 8.34 / 10*

Robotics & Automation Minor (GPA: **9.14**)
Relevant Coursework: Control Systems, Modern Control Systems, Industrial Instrumentation and Control, Robotics & AI for Robotics, Computer Architecture, Analog Electronics, Machine Learning.

Research Projects

Hybrid Control for Dual Arm Systems (PoppyRL), (SoTeleIK) **Feb 2025 – Present**
Supervised by Prof. Arshad Javed & Prof. Abhishek Sarkar

- Deployed a **Levenberg–Marquardt** based inverse kinematics solver for accurate and stable teleoperation tracking.
- Developed a reinforcement learning framework for dual-arm inverse kinematics in humanoid robots and implemented continuous-control algorithms **(SAC, DDPG, HER)** to enable object manipulation
- Technologies: Reinforcement Learning, SAC, DDPG, HER, LM-IK, CoppeliaSim, PyBullet, Python.

Evolutionary Controllers for Planar Snake Bots (Sir-penoid) **Jan 2024 – April 2025**
Supervised by Prof. Joyjit Mukherjee & Prof. Abhijit Das

- Designed and trained adaptive neural-network-based controllers using evolutionary genetic algorithms to achieve highly flexible snake-like locomotion across diverse flat environments.
- Achieved a **30% faster routing time and a 73% more memory-efficient controller** compared to the current algorithm.
- Work submitted as **first author to ICRA 2025; awaiting review**
- Technologies: Python, PyBullet, Dynamixel, Neural Networks, Genetic Algorithms.

Design and Control of an Autonomous Bicopter **Aug 2023 – Jan 2024**
Supervised by Prof. Joyjit Mukherjee & Prof. Abhishek Sarkar

- Developed a **Lyapunov-based nonlinear control approach** to stabilize a complex underactuated bi-copter platform.
- ensured robust trajectory with **tracking precision of upto 1cm and disturbance rejection.**
- Developed and verified the controller in Gazebo, **currently constructing a HIL prototype for hardware validation.**
- Technologies: Python, Simulink, ROS, Gazebo, Raspberry Pi, Teensy.

Selected Projects & Competitions

Hierarchical Control Autonomy Framework (H-CAF) for Drones (H-CAF) **May 2025 – Present**
Technologies: PX4, OpenCV, QGC, Mavlink, ROS2, Python

- Designed and developed H-CAF, a modular and hierarchical drone autonomy framework integrating feedback linearization (FBL) with linear parameter-varying model predictive control (LPV-MPC) to achieve robust trajectory tracking and improved disturbance rejection.
- Implemented H-CAF in PX4-ROS2 simulation-in-the-loop (SITL) environments, enabling seamless integration between high-level mission planning modules and low-level flight controllers.
- Planning to develop vision-based perception pipelines using OpenCV for real-time waypoint detection, dynamic obstacle avoidance, and enhanced situational awareness.
- Aiming to lay the foundation for fully autonomous, vision-guided flight missions capable of adapting to changing environments and mission objectives.

LAM Research Challenges 2024 **Oct 2024 – Jan 2025**
Technologies used: Arduino, basic analog/digital electronics, PID control, standard libraries

- Qualified to the second round of the highly competitive LAM Research Challenges, among top teams selected nationally.
- Designed and implemented a PID controller from scratch to precisely regulate a thermal system's temperature under dynamic load conditions, using only standard Arduino libraries and discrete electronic components.
- Engineered the system to achieve stability, and minimal steady-state error, demonstrating deep understanding of control theory and practical system limitations.
- Collaborated as part of a multidisciplinary team, managing electronics design, controller tuning, and integration within strict hardware constraints set by the competition.

IRoC-U – 2024 **Nov 2023 – Jun 2024**
Technologies: ROS2, Python, C++, OpenCV, SLAM, Nav2, LoRa, Raspberry Pi

- Collaborated as part of a 10-member team to engineer a fully autonomous rover with rocker-bogie suspension for ISRO's Robotic Challenge 2024.
- Simulated the robot in a custom ROS2-Gazebo environment provided by the organizers, validating motion planning and perception algorithms under realistic terrain and task constraints.
- Implemented SLAM for real-time mapping and localization; integrated with the Nav2 stack for autonomous waypoint-based navigation and obstacle avoidance.

- Used ROS2 Control to model and simulate motor control, enabling seamless transition from simulation to real-world deployment.
- Built the physical rover using microcontrollers and a Raspberry Pi, linked to a remote base station via long-range LoRa communication for command and telemetry.
- Successfully progressed to the penultimate round of the competition, demonstrating strong system integration, autonomy, and mission planning capabilities.

SAE AEROTHON 2023 **Feb – Jun 2023**
Technologies: PX4, OpenCV, QGC, MAVLink, Python

- Qualified for the presentation round as one of the few first-year teams nationally.
- Tasked with visually detecting circular waypoints, estimating their GPS coordinates using onboard camera feeds and current drone GPS location.
- Implemented OpenCV-based Hough Circle Transform for robust detection and localization of waypoints in real-world conditions.
- Planned an optimal waypoint path using graph-based node traversal algorithms; transmitted navigation goals to PX4 via MAVLink.
- Executed autonomous landing by detecting concentric landing targets in real-time and issuing precise landing commands aligned with the center of the target.

Experience

Research Intern (FlyingArm) **Aug – Present**
RRC - IIIT hyderbad Hybrid

- Developed a **unified aerial-manipulation platform with a quadrotor and lightweight robotic arm**, including controller design, hardware integration, and modular ROS2/PX4 interfaces.
- Applied **DroneDiffusion's** diffusion-based residual-dynamics controller to provide noise-free trajectory tracking and mimic complex quadrotor-manipulator disturbances.
- Used reinforcement learning (PPO, SAC) **to simulate agile flight behaviors**, including high speed waypoint tracking.
- Currently trying to **integrate Pegasus simulator with NVIDIA Isaac Lab** to create a sim-to-real pipeline for deploying learnt policies to actual aerial-manipulators.

Research Intern (BeetleSim) **May – Aug 2024**
AITG group – CSIR-CEERI, Pilani On-site

- Designed and built a fully autonomous rover with real-time **SLAM-based mapping and localization, integrated with the Nav2 stack** for robust waypoint navigation and global planning, and experimented with Artificial Potential Fields (APF) to enhance local obstacle avoidance in dynamic environments.
- Developed a **neuroevolution-based control strategy** combining genetic algorithms and neural networks to enhance locomotion capabilities, improving adaptability to uneven and unpredictable terrains.
- Engineered a custom Betaflight-compatible flight controller PCB for coordinated multi-drone swarm operations.

Teaching Assistant (Control Systems) **Jan – May 2025**
BITS Pilani, Hyderabad On-site

- Collaborated with Professor Joyjit Mukherjee to assist students in understanding and solving Control Systems problems during tutorial sessions.
- Developed a comprehensive question bank by curating and creating practice problems focused on time-domain and frequency-domain analysis, root locus, and state-space methods.
- Served as the primary point of contact for students seeking clarification on complex topics such as stability analysis, compensator design, and modern control concepts, providing targeted guidance and mentorship.

Technical Skills and Interests

Languages: Python, C/C++, MATLAB
Libraries & Frameworks: Tensorflow, PyTorch, Gym, NumPy, OpenCV, Matplotlib, MATLAB/Simulink Block sets
Simulation, Control & Robotics Tools: ROS2, PX4, Gazebo, MuJoCo, Isaacs CoppeliaSim, Simulink
Areas of Interest: Control, Reinforcement Learning, Machine learning, UAV, Manipulators, Robotics
Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability, Resilience under pressure, Curiosity

Positions of Responsibility

President : Automation and Robotics Club, BITS Pilani, Hyderabad **May 2024 – 2025**

- Led one of the largest and oldest robotics clubs on campus, mentoring over 60 active members and fostering a strong technical and collaborative culture.
- Strategically increased the club's annual project output from 1 to over 5 recognized and impactful robotics and automation projects per year.
- Conceptualized and organized workshops, hands-on sessions, and hackathons to enhance robotics literacy and practical skills across the student community.
- Initiated outreach programs in collaboration with "Nirmaan" to conduct robotics workshops and STEM awareness sessions for students from underprivileged backgrounds.
- Acted as the primary representative for the club in institute-level meetings, competitions, and external collaborations, strengthening its campus and external presence.